Experiment 9 - Experiment to understand Matplotlib library use cases in Data Science through visualization.

Aim:

To conduct an experiment to understand Matplotlib library use cases in Data Science through visualization.

Program:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

%matplotlib inline
data = pd.read_csv('/content/drive/MyDrive/Iris_Dataset.csv')
data
data.info()
data.describe()

data.value_counts('variety')
sns.countplot(x='variety', data=data,)
plt.show()

dummies = pd.get_dummies(data.variety)
FinalDataset = pd.concat([pd.get_dummies(data.variety), data.iloc[:, [0, 1, 2, 3]]], axis=1)
FinalDataset.head()

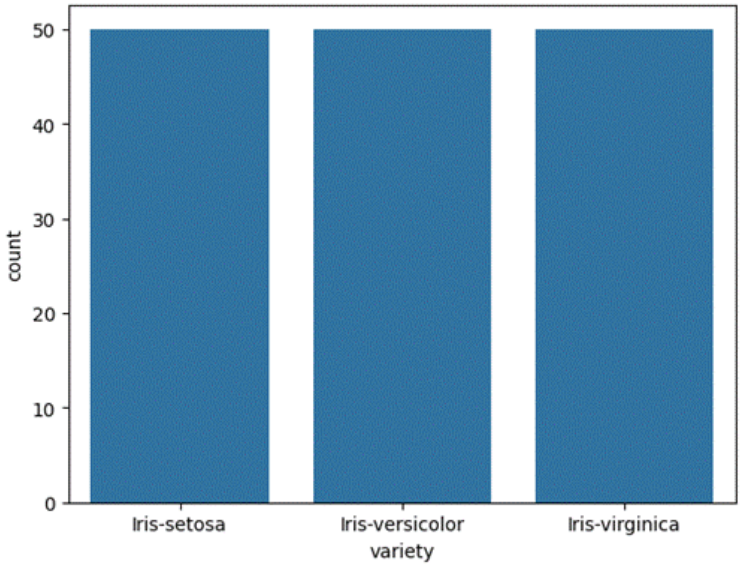
sns.scatterplot(x='SepalLengthCm', y='SepalWidthCm', hue='variety', data=data,)
sns.scatterplot(x='PetalLengthCm', y='PetalWidthCm', hue='variety', data=data,)

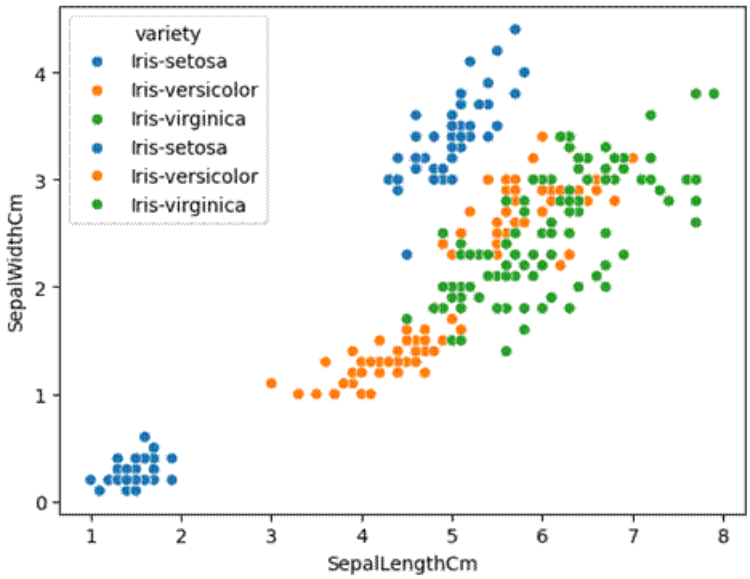
sns.pairplot(data, hue='variety', height=3);
plt.show()
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, 'PetalLengthCm').add_legend();
plt.show();
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, 'PetalWidthCm').add_legend();
plt.show();
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, 'SepalLengthCm').add_legend();
plt.show();
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, 'SepalWidthCm').add_legend();
```

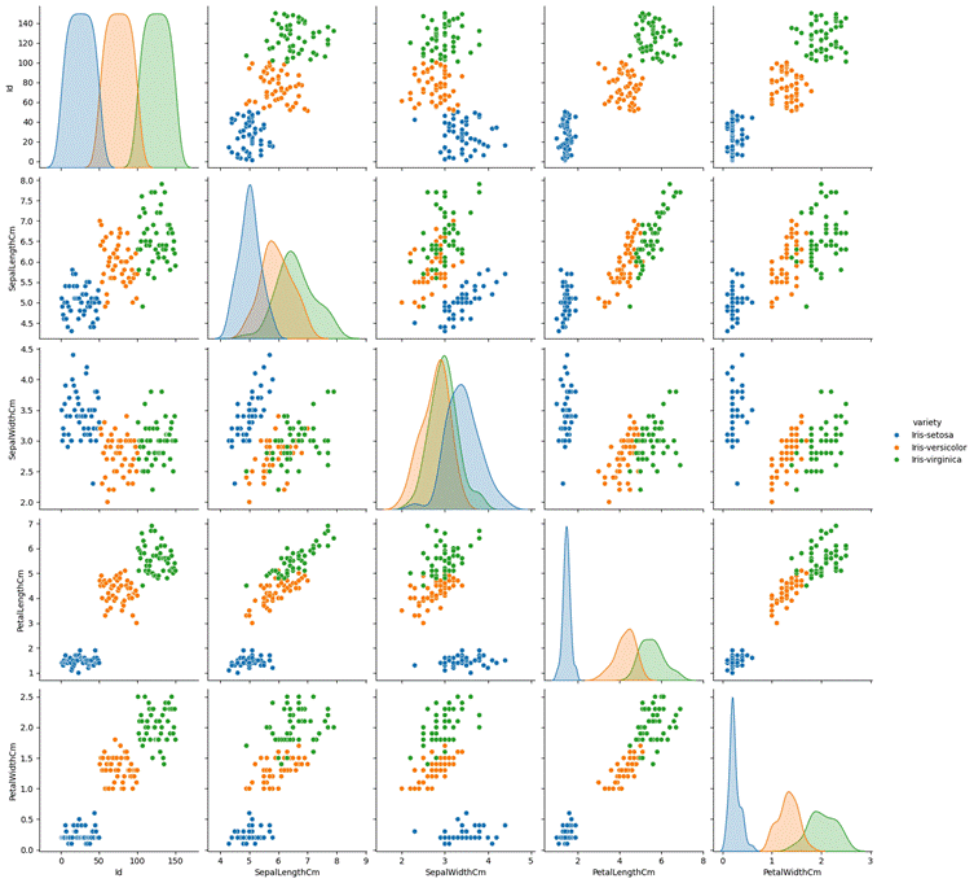
plt.show();

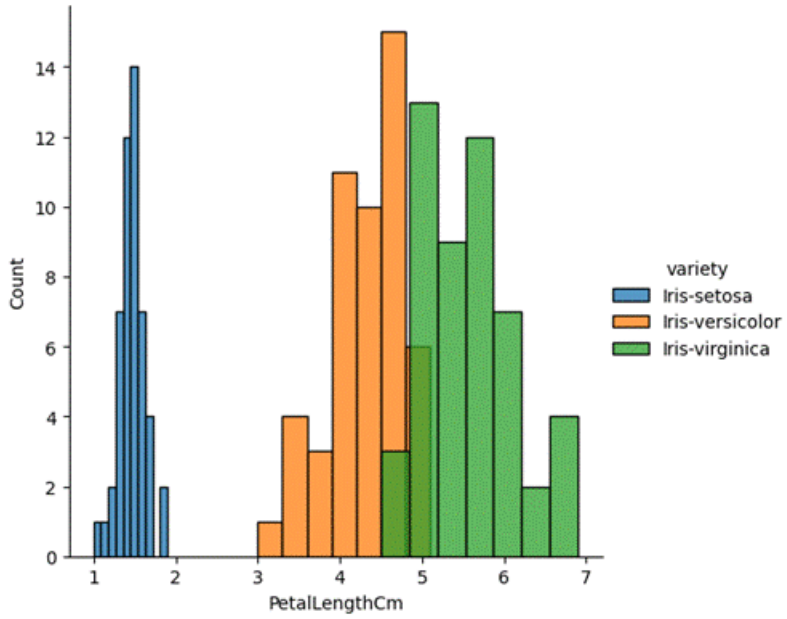
Output:

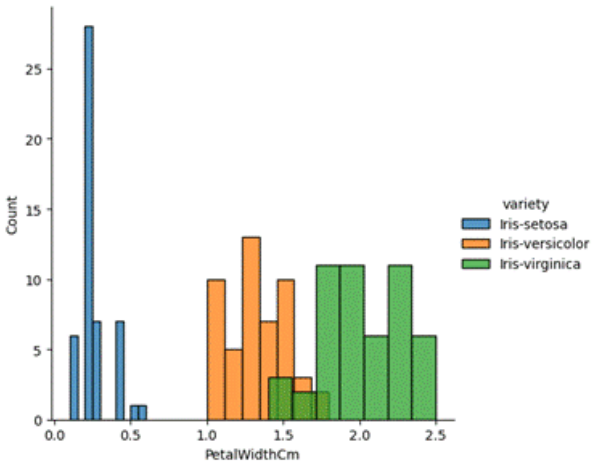
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

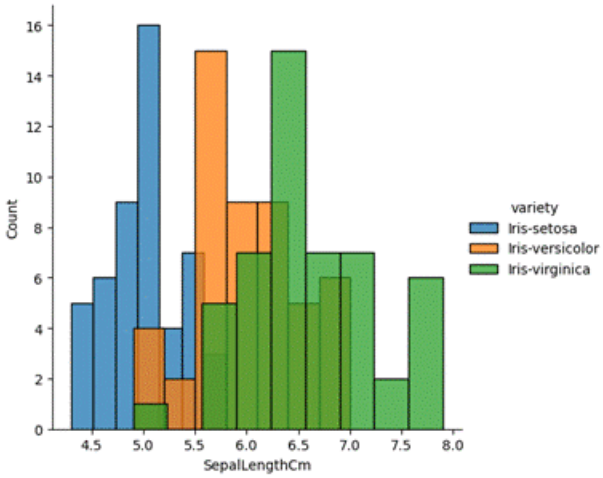


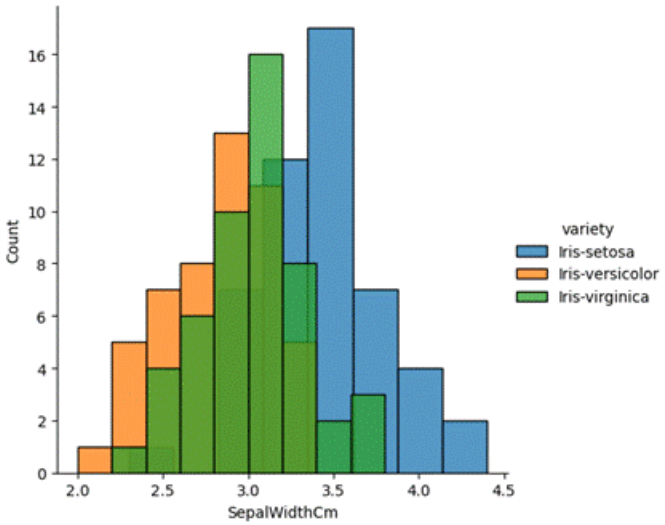



Id
SepalLengthCm
SepalWidthCm
PetalLengthCm
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
PetalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica





Result:

Thus, an experiment to understand Matplotlib library use cases in Data Science through visualization was conducted.